

● HARD

● INFORMATION AND IDEAS

Central Ideas and Details

03

10 questions · ~15 min

Q1

INFORMATION AND IDEAS

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

- A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.
- B. Fynbos roots extend close enough to the forest plants' roots that they constitute a physical barrier that forest plants' roots cannot pass.
- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.
- D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.

Optical tweezers are specialized scientific tools—particularly useful in biology and medicine—that use high-powered beams of light to trap and manipulate minuscule particles for study. Use of the tool has led to several scientific and medical breakthroughs over the last few decades, but the particles are often under prolonged exposure to the intense heat of the light beams. To overcome the risk of overheating, and thereby damage, researchers sometimes attach nano-sized glass beads to particles, allowing the light to focus on the beads instead of the particles.

Based on the text, what is one advantage of attaching glass beads to particles when using optical tweezers?

- A. It decreases the time it takes for the optical tweezers to locate and capture the particles.
- B. It facilitates the maneuvering of particles without directly heating the particles themselves.
- C. It allows researchers to use weaker light beams to manipulate particles.
- D. It adds a material to which particles can transfer any heat absorbed from the optical tweezers' light beam.

The following text is from Milan Kundera's 1984 novel *The Unbearable Lightness of Being* (translated by Michael Henry Heim in 1984). Karenin is a dog that belongs to Tomas and Tereza.

Karenin was not overjoyed by the move to Switzerland [from Prague]. Karenin hated change. Dog time cannot be plotted along a straight line; it does not move on and on, from one thing to the next. It moves in a circle like the hands of a clock, which—they, too, unwilling to dash madly ahead—turn round and round the face, day in and day out following the same path. In Prague, when Tomas and Tereza bought a new chair or moved a flower pot, Karenin would look on in displeasure. It disturbed his sense of time. It was as though they were trying to dupe the hands of the clock by changing the numbers on its face.

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Which choice best states the main idea of the text?

- A. As a dog, Karenin possesses a sense of time that involves a strong preference for predictability and an aversion to disruption.
- B. After he's moved to a new home, Karenin's negative response to changes has become more pronounced.
- C. Similar to Tomas and Tereza, Karenin comprehends time as circular rather than as a straightforward progression.
- D. As is the case for other dogs, Karenin's sense of time seems to accelerate depending on the objects and places that surround him.

Eighteenth-century economist Adam Smith is famed for his metaphor of the invisible hand, which he putatively used to illustrate a robust model of how individuals produce aggregate benefits by pursuing their own economic interests. Note “putatively”: as Gavin Kennedy has shown, Smith deploys this metaphor only once in his economic writings—to make a narrow point about the then-dominant economic theory of mercantilism—and it was largely ignored until some twentieth-century economists eager to secure an intellectual pedigree for their views elevated it to a fully-fledged paradigm.

Which choice best states the main idea of the text?

- A. Although Smith is famed for his metaphor of the invisible hand, the metaphor was largely ignored until economists in the twentieth century came to realize that the metaphor was a robust model that anticipated their own views.
- B. Some twentieth-century economists gave Smith’s metaphor of the invisible hand a significance it does not have in Smith’s work, but it is nevertheless a useful model of how individuals produce aggregate benefits by pursuing their own economic interests.
- C. Smith’s metaphor of the invisible hand has been interpreted as a model of how individuals acting in their own interest produce aggregate benefits, but it was intended as a subtle critique of the economic theory of mercantilism.
- D. The reputation of Smith’s metaphor of the invisible hand is not due to the importance of the metaphor in Smith’s work but rather to the promotion of the metaphor by some later economists for their own ends.

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH_3 (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH_3 likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH_3 .

Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH_3 ?

- A. Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.
- B. Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.
- C. It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.
- D. It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

- A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum's long-suspected role in motor coordination.
- B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.
- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
- D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

- A. It is the earliest known example of a titanosaur nesting and breeding site.
- B. It was very difficult to excavate given that it was discovered in a mine.
- C. It may have been occupied by other sauropods in addition to titanosaurs.
- D. It is farther north than any other nesting site discovered in South America.

The following text is from William Shakespeare's play *The Tempest*, first performed in 1611. Miranda has lived on an island with her father, Prospero, since she was three years old. Prospero has stated that Miranda likely does not remember anything other than her life on the island.

MIRANDA: 'Tis far off,
And rather like a dream than an assurance
That my remembrance warrants. Had I not
Four or five women once that tended me?

PROSPERO: Thou hadst, and more, Miranda. But how is it
That this lives in thy mind? What seest thou else
In the dark backward and abysm of time?
If thou remember'st ought ere thou camest here,
How thou camest here thou mayst.

In the text, which point does Prospero most directly make about Miranda and her memories?

- A. Miranda's reminiscences about her early childhood have a melancholy quality that betrays her discontented view of her current circumstances.
- B. Miranda's doubts about the accuracy of one recollection of a place other than the island are clouding her judgment and seem to be making her reluctant to explore her recollection of traveling to the island.
- C. Miranda's ability to summon details of an experience she had before arriving on the island suggests that she may also be able to summon details of her arrival on the island.

- D. Miranda's impression of a scene is vague because she is remembering a scenario she had daydreamed about as a child rather than a scenario that had occurred in reality.

Some animal-behavior studies involve observing wild animals in their natural habitat, and some involve capturing wild animals and observing them in a laboratory. Each approach has advantages over the other. In wild studies, researchers can more easily presume that the animals are behaving normally, and in lab studies, researchers can more easily control factors that might affect the results. But if, for example, the results from a wild study and a lab study of Western scrub-jays (*Aphelocoma californica*) contradict each other, one or both of the studies must have failed to account for some factor that was relevant to the birds' behavior.

Which choice best states the main idea of the text?

- A. When the results of a natural-habitat study and those from a lab study of a wild animal such as the Western scrub-jay conflict, the study in the natural habitat is more likely than the lab study to have accurate results.
- B. Studying wild animals such as the Western scrub-jay in both their natural habitat and lab settings is likely to yield conflicting results that researchers cannot fully resolve.
- C. Wild animals such as the Western scrub-jay can be effectively studied in their natural habitat and in the lab, but each approach has drawbacks that could affect the accuracy of the findings.
- D. Differing results between natural-habitat and lab studies of wild animals such as the Western scrub-jay are a strong indication that both of the studies had design flaws that affected the accuracy of their results.

Poetry in Classical Nahuatl, the language of the Aztec Empire, relies on *difrasismo*, or a parallel noun construction that conventionally operates as a single metaphor. For example, the common difrasismo *in cuauhtli in ocelotl* (literally, “the eagle, the jaguar”) signifies “warrior.” The device’s function is both formal—providing structure to lines of verse—and ritual: semantic relations among the two nouns and the concept they signify can be tenuous, as in the previous example, such that difrasismos are often only intelligible according to the conceptual associations observed in Aztec ceremonial culture.

Which statement about the difrasismo *in cuauhtli in ocelotl* is most strongly supported by the text?

- A. Its metaphorical significance derives from the semantic equivalence of the two nouns constituting the difrasismo.
- B. Its unintelligibility may cause its formal function within a line of verse to go unnoticed by present-day readers.
- C. Its apparent obscurity can be resolved when considered in the proper cultural context.
- D. Its frequency in Classical Nahuatl poetry confirms its intelligibility to the Aztec audience.